Answers and talk notes

Represent Rational Numbers on the Coordinate Plane

What your student is learning

Tonight we reflect points across the x-axis and the y-axis — and see exactly which coordinate changes sign.

Answers

Ask your student to explain before you look.

1. **B.** The y-coordinate. The y-coordinate changes sign across the x-axis, and x stays the same.
2. **B.** (4, −3). Across the x-axis, y changes sign: (4, −3).
3. **D.** (−6, 1). Across the y-axis, x changes sign: (−6, 1).
4. **A.** (−5, −2). Across the x-axis, y changes sign: (−5, −2).
5. **C.** (−1.5, −4). Across the y-axis, x changes sign: (−1.5, −4).

Talk about it

Which coordinate changes sign when you reflect across the x-axis? Which stays the same?

Home mission

Lay a ruler on the table as the x-axis. Put a coin below it for the mural star. Ask: where does its mirror copy go? Put a second coin there. Then compare how far each coin is from the ruler. Which number changed, the left-right one or the up-down one?

When they are stuck, ask

- Which axis are we reflecting across?
- Does that axis run left-right or up-down? Which value has to change?
- Is the image the same distance from the mirror line?

Try not to say

- Reflecting over the x-axis changes the x-coordinate.
- Flip both coordinates, it is safer.
- The reflected point can be any distance from the axis.